```
ansible/inventory/kvm/
       /playbooks/configuration/
       /files/
```

The *inventory/kvm/inventory* file contains the following:

```
debian ansible_host=192.168.122.140 ansible_connection=ssh
ansible_user=debian
```

The *ansible/files/* directory has the following:

```
ansible/files/_site
             /example.domain
             /goaccessrc
```

The default Debian 9 installation does not install the sudo package. Start the new Debian VM, log in as the root user, and install the sudo package. You should also provide sudo access to the user account, which is 'debian' (in this example).

```
root@debian:~# apt-get install sudo

root@debian:~# adduser debian sudo
Adding user `debian' to group `sudo'...
Adding user debian to group sudo
Done.
```

You should now be able to issue commands from the host system to the guest OS, using Ansible. For example:

```
$ ansible -i inventory/kvm/inventory debian -m ping

debian | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

Nginx

The first step is to install the Nginx Web server. The software package repository is updated, and then the Nginx package is installed. The Nginx Web server is started and we wait for the server to listen on port 80. The playbook to install Nginx is as follows:

```
---
- name: Install nginx
  hosts: debian
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [nginx]

  tasks:
    - name: Update the software package repository
      apt:
        update_cache: yes

    - name: Install nginx
      package:
        name: "{{ item }}"
        state: latest
      with_items:
        - nginx

    - name: Start nginx
      service:
        name: nginx
        state: started

    - wait_for:
        port: 80
```

The above playbook can be invoked using the command shown below:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/nginx.yml --tags nginx -vv -K
```

The *-vv* represents the verbosity in the Ansible output. You can use up to four *v*'s for a more detailed output. The *-K* option prompts for the sudo password for the *debian* user account. You can now open a Web browser on the local host with the URL *http://192.168.122.140* and you should see the default Nginx home page as shown in Figure 1.

Figure 1: Nginx default home page

Serving static files

You can use any static site generator to create your HTML, CSS and JavaScript files. You will need to copy the generated files to the *files/_site* directory. The Ansible playbook to copy the static files to the Nginx Web server location is given below:

```
- name: Copy static files
  hosts: debian
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [static]
```